# CHAPTER 9

## STATISTICAL ANALYSIS OF THE INHIBITORY EFFECTS OF LACTOFERRIN AGAINST MULTI-DRUG RESISTANT PATHOGENS

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## INTRODUCTION

The transferrin family of proteins includes lactoferrins, which bind iron. Since their initial isolation from both bovine and human milk, lactoferrins have undergone extensive structural and functional investigations. This is particularly true given their wide range of functions, characteristics, and uses in the food and pharmaceutical industries (Johansson et al. 1960, Groves et al. 1960).

Although lactoferrins have been found in other mammalian species, bovine and human lactoferrins have received the most attention thus far. Colostrum, milk, tears, nasal and bronchial secretions, saliva, bile and pancreatic secretions (and thus in gastric and intestinal fluids), urine, and seminal and vaginal fluids are just a few exocrine secretions that contain lactoferrins because they are produced by epithelial cells of various body organs (Masson et al. 1971, Heremans et al. 1971). Additionally, the hematopoietic tissue of the bone marrow also produces lactoferrins, which are present in the granules of polymorphonuclear neutrophils (Berlov et al. 2007).

The most plentiful sources of lactoferrins are colostrum and milk, whose quantities vary depending on the type of mammal and lactation stage. Other exocrine secretions have lower lactoferrin amounts. Additionally, different mammalian species have different quantities of lactoferrin in neutrophils (Barton et al. 1988). Despite the fact that lactoferrin from neutrophils in humans has the same structure as lactoferrin from milk (Moguilevsky et al. 1985).

With a molecular weight of about 80 kDa, lactoferrins are monomeric glycoproteins with well-known three-dimensional geometries and amino acid sequences. There have been reports on the primary amino-acid sequences of lactoferrins from a variety of mammalian species, including those for the human (Moore et al. 1997, Cutone et al. 2020), cow (Pierce et al. 1991), goat (Lee et al. 1997), sheep, camel (Khan et al. 2001), buffalo (Karthikeyan et al. 1999), horse (Ashwan et al.1998). Human lactoferrin has two extra amino acids (691 total), compared to bovine lactoferrins single polypeptide chain of 689 amino acids (Lepanto et al. 2019).

Since lactoferrins have a lot of promise as a natural defense agent, antibacterial activity was the first of these to be attributed to them (Zarzosa et al. 2020). In addition to having a wide range of antibacterial activities against Gram-positive and Gram-negative bacteria and having the potential to be used as natural antibiotics in human and veterinary medicine, lactoferrin and lactoferrin-derived peptides also have a wide range of antiviral activities against enveloped and naked viruses (Redwan et al. 2014), fungi, yeast, and parasites (Fernandes et al. 2017, Leboffe et al. 2009, Giansanti et al. 2009).

Additionally, lactoferrin has been touted as an effective medication for the current COVID-19 pandemic brought on by the severe acute respiratory syndrome coronavirus 2. (SARS-CoV2). Although lactoferrin also exhibits antimicrobial activity in the iron-independent route through direct interaction with microbes, its antimicrobial activity is tied to its capacity to chelate iron and so deprive germs of these vital nutrients (Wang et al. 2020, Elnagdy et al. 2020, Zimecki et al. 2021).

## Production

Different ion exchangers were investigated by (Stanic et al. 2012) for the segregation of whey proteins. In this work, numerous biochemical techniques to separate lactoferrin from milk of diverse species have been addressed. There have been attempts to isolate lactoferrin from goat, elephant, sheep, alpaca, camel, gray seal, and human milk. An SP-Sepharose is used to purify lactoferrin utilizing cation exchange chromatography. SDS-PAGE with 10% polyacrylamide gel is used to identify the lactoferrin that was isolated by chromatography. It is then freeze-dried and kept at 20°C for storage. Use of the optimal wavelength, 280 nm, for ion exchange chromatography using a monolithic column attached with a UV-VIS detector is possible. The flow rates of the retention and elution solutions are 0.25 ml/min and 0.75 ml/min, respectively, and the ionic strength of the elution solution is 1.5 M NaCl. Under ideal circumstances, the detection limit of lactoferrin was 0.1 g/ml.

***Pseudomonas aeruginosa***

The Gram-negative bacteria *Pseudomonas aeruginosa*, which is typically prevalent in hospitals, is essential for nosocomial infection as well as acute and chronic infection. *P. aeruginosa* is widely distributed in nature and exhibits a high resistance to many antibiotic classes (Kerckhoffs et al. 2011). The bacteria quickly proliferate, colonize surfaces with water, and perform all of their metabolic processes for growth and development, forming an association of complex matrix called a biofilm (Murga et al. 2001).

According to the study, a biofilm renders bacteria more vulnerable to factors like antibiotic use, UV radiation exposure, and salt (Johani et al. 2018). A wide range of research is being done to better understand the causes and mechanisms of resistance. *Pseudomonas* species exhibit high resistance to many kinds of antibiotics that are employed to persistently eradicate the microbial infection because of the complicated biofilm-forming capabilities. The presence of Pseudomonas species in hospitals aids in the formation of biofilms on medical devices, including implants in patients and other similar devices with exposed surfaces (De Silva et al. 2017).

For the study of the biochemical pathways underlying the pathovars' susceptibility to numerous antibiotics groups, including amikacin, gentamicin, carbapenem, ofloxacin, ciprofloxacin, tigecycline, tobramycin, and norfloxacin, *Pseudomonas* species are employed as a model organism (Mesaros et al. 2007, Moore et al. 2016).

By modifying the genome sequences and the expression of proteins that ultimately strengthen the resistance of the pathovars, the pathogenic *Pseudomonas* species create a significant problem in the diversity of bacteria (Overhage et al. 2008). Due to the bacteria's resistance, numerous metabolic processes and channel protein functions are impacted (Tenover 2006, Miko et al. 2019).

**Materials and methods of work**

In the current study, high-efficiency devices such as a pressure oven, a digital camera, a sensitive balance, a deep freeze, and an optical microscope are used from international origins.

**Sample Collection**

In the period from February to May 2022, children and adults (outpatient and inpatient) at Azadi Hospital in Kirkuk city participated in the study. A total of (71) patient agreed to participate, and an interview was conducted for each patient under the supervision of a physician. Various clinical samples (wound, blood, urine, sputum) will be produced.

**Effect of lactoferrin on bacterial growth**

100 L of 24-hour-old cultures will be transferred to 96-microtitre plates with MIC contraction of lactoferrin or lactoferrin with a chosen antibiotic after 24-hour-old cultures are diluted to an initial OD600nm 0.1. Afterward, the culture plate will incubate for 24 hours at 37°C. Spectrophotometric growth monitoring (OD600nm) will be done after 2, 4, 6, 8 and 24 hours of incubation at 37°C. There will also be a bad control. For each test, three separate experiments will be conducted.

**Identification**

The shape, diameter and shapes of the bacterial isolates were determined on blood agar, mannitol agar, chocolate agar and McConkey agar. Microscopic and biochemical examinations were used, which included qualitative tests for each type, as well as the System Compact Vitek-2 device, to confirm the results of the biochemical determination. The results of Vitek-2 were identical to the biochemical tests and the proportions of bacteria isolated from the clinical sources were determined as follows:

**Table 1.** Numbers and percentage of *Pseudomonas aeruginosa* isolated from clinical source

| Bacteria isolates | Urine | Wound | Blood | Sputum | Total | C.S. (*) *P-value* |
|---|---|---|---|---|---|---|
| ***P. aeruginosa*** | 15(22.4%) | 19(27.9%) | 5(45.5%) | 3(11.5%) | 42(24.4%) | C.C.=0.071 P=0.973 (NS) |
| ***A. baumonii*** | 1(1.5%) | 3(4.4%) | 3(27.3%) | 6(23.1%) | 13(7.6%) | C.C.=0.251 P=0.150 (NS) |
| ***S. aureus*** | 40(59.7%) | 25(36.8%) | 1(9.1%) | 12(46.2%) | 78(45.3%) | C.C.=0.184 P=0.172 (NS) |
| ***S.pyogenus*** | 11(16.4%) | 21(30.9%) | 2(18.1%) | 5(19.2%) | 39(22.7%) | C.C.=0.067 P=0.797 (NS) |
| **Total** | 67(38.9%) | 68(39.6%) | 11(6.4%) | 26(15.1%) | 172(100%) | C.C.=0.000 P=1.000 (NS) |

### Identification of *Pseudomonas aeruginosa*

Colony morphology of *Pseudomonas aeruginosa* in MacConkey agar showing 2-3 mm, flat, smooth, non-lactose fermenting colonies with regular margin and Alligator skin like from top view (Figure 1).

**Figure 1.** *P. aeruginosa* colonies on MacConkey agar

The colorless colonies on MacConkey agar and the huge flat colonies that developed zones of beta-haemolysis with a grape-like odor on the latter media were grown onto Cetramide agar. Colonies had a yellow-blue/green pigmentation on this medium. Therefore, based on the data, we can infer that these isolates may be Pseudomonas species. When grown on cetramide agar, which serves as a selective differential medium for the identification of *P. aeruginosa* and functions as a detergent and inhibits the growth of other microbes, the ability to create green-blue/yellow pigments is demonstrated. Pyocyanin and a brilliant yellow-green pigment are produced by this organism in response to the iron level of the media.

**Table 2.** Biochemical tests of gram negative isolated.

| Biochemical tests Types | | Catalase | Kliguler test | Motility | $H_2S$ | Simmons citrate | V-P | M-R | Indol | Oxidase |
|---|---|---|---|---|---|---|---|---|---|---|
| *P. aeruginosa* | S1 | + | - | + | - | - | + | - | - | + |
| | S2 | + | - | + | - | - | + | - | - | + |
| | S3 | + | - | + | - | - | + | - | - | + |
| | S4 | + | - | + | - | - | + | - | - | + |
| | S5 | + | - | + | - | - | + | - | - | + |
| *A. baumannii* | S1 | + | - | - | - | + | - | + | - | - |
| | S2 | + | - | - | - | + | - | + | - | - |
| | S3 | + | - | - | - | + | - | + | - | - |
| | S4 | + | - | - | - | + | - | + | - | - |
| | S5 | + | - | - | - | + | - | + | - | - |

**Table 3:** Biochemical tests of gram positive isolated.

| Biochemical tests / Types | | Motility | Hemolysis | Urease test | Mannitol salt agar | Novobiocin antibiotic | Coagulase | Oxidase | Catalase | Gram stain |
|---|---|---|---|---|---|---|---|---|---|---|
| *S.aureus* | S1 | - | B | + | + | / | + | - | + | + |
| | S2 | - | B | + | + | / | + | - | + | + |
| | S3 | - | B | + | + | / | + | - | + | + |
| | S4 | - | B | + | + | / | + | - | + | + |
| | S5 | - | B | + | + | / | + | - | + | + |
| *S.pyogenes* | S1 | - | B | - | - | - | - | - | - | + |
| | S2 | - | B | - | - | - | - | - | - | + |
| | S3 | - | B | - | - | - | - | - | - | + |
| | S4 | - | B | - | - | - | - | - | - | + |
| | S5 | - | B | - | - | - | - | - | - | + |

**Antibiotic susceptibility test for *Pseudomonas aeruginosa***

Amoxicillin/clavulanic acid, ampicillin, tetracycline, cefoxitin, ceftriaxone, chloramphenicol, and trimethoprim were all completely ineffective against *P. aeruginosa*. 50% resistance to Amikacin, 75% resistance to Nitrofurantoin and Nalidixic Acid. 25% had Doxycycline, Azithromycin, and Ciprofloxacin resistance. Tobramycin, Cefepime, and Imipenem all demonstrated complete sensitivity to P. aeruginosa, respectively. Contrarily, 75% are susceptible to azithromycin and gentamicin. According to the table, 50% of people are susceptible to ciprofloxacin, doxycycline, and amikacin (Table 4.5).

The new findings are in line with Al-Jendy and Al-Ofairi's (2019) claim that *P. aeruginosa* was 100% resistant to Amoxicillin and Penicillin in terms of antibiotic resistance. Other than that, Ciprofloxacin has little effect on *Pseudomonas aeruginosa*. Prior reports indicated that Ciprofloxacin was highly effective against UTIs.

**Table 4.** Antibiotic susceptibility test of *P. aeruginosa*.

| Antibiotics | Resistant | | Sensitive | | Intermediate | |
|---|---|---|---|---|---|---|
| | No. | % | No. | % | No. | % |
| **AMP (10mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **AMC (10/20mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **CX (30mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **CRO (30mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **CPM (30mg)** | 0 | 0 | 4 | 100 | 0 | 0 |
| **IMI (10mg)** | 0 | 0 | 4 | 100 | 0 | 0 |
| **CN (10mg)** | 0 | 0 | 3 | 75 | 1 | 25 |
| **AK (30mg)** | 2 | 50 | 2 | 50 | 0 | 0 |
| **TE (30mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **DXT (30mg)** | 1 | 25 | 2 | 50 | 1 | 25 |
| **C (30mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **AZM (15mg)** | 1 | 25 | 3 | 75 | 0 | 0 |
| **NA (30mg)** | 3 | 75 | 1 | 25 | 0 | 0 |
| **CIP (5mg)** | 1 | 25 | 2 | 50 | 1 | 25 |
| **TMP (5mg)** | 4 | 100 | 0 | 0 | 0 | 0 |
| **F (300mg)** | 3 | 75 | 1 | 25 | 0 | 0 |
| **TOB (10mg)** | 0 | 0 | 4 | 100 | 0 | 0 |

R=Resistant, S=Sensitive, I=Intermediate, AMP= Ampicillin, AMC= (Amoxicillin and Clavulanic acid (Augmentin)), CRO=Ceftriaxone, CPM= Cefepime, IMI= Imipenem, CN=Gentamicin, AK=Amikacin, TE=Tetracycline, DXT= Doxycyclin, C=Chloramphenicol, AZM=Azithromycin, NA=Nalidixic acid, CIP=Ciprofloxacin, TMP=Trimethoprim, F=Nitrofurantoin, TOB= Tobramycin, CX=Cefoxitin

According to a study by (Al-Salamy et al, 2012. Emad et al, 2012), the susceptibility of 181 clinical isolates of *P. aeruginosa* showed that ampicillin had the highest resistance (100%) in accordance with the current findings. Cephalothin (92.81%), Cefotaxem (84.53%), Gentamycin (69.61%), and Trimethoprim (67.99%) were the next most resistant antibiotics. Tobramycin (14.91%) and Amikacin (14.3%) are two of the antibiotics with the lowest resistance rates that are most effective against *P. aeruginosa*.

They found that the majority of P. aeruginosa isolates were susceptible to amikacin (79%), tobramycin (70%) and piperacillin (65%), in contrast to the current findings (Anjum et al. 2010, Mir et al.

2010). There are many factors that contribute to the spread of *P. aeruginosa* resistance to the majority of antibiotics in humans and animals. Reducing unnecessary antibiotic use, treating with narrow spectrum agents, improving adherence to therapy, reducing antibiotic use in animals and agriculture, and improving infection control are all important ways to address this issue. Additionally, vaccination can lessen the effects of resistance by reducing the risk of infection and transmission.

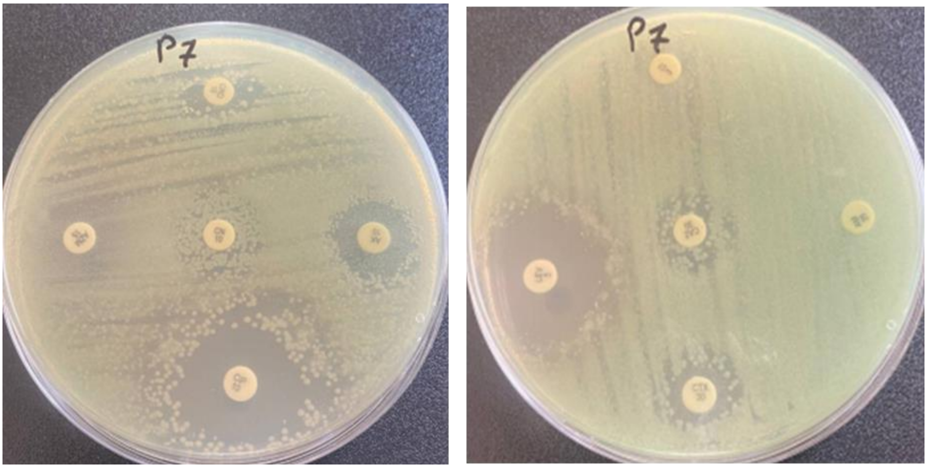


**Figure 2.** Antibiotic susceptibility test of *P. aeruginosa*

The antimicrobial activity of lactoferrin against the studied pathogenic bacteria was investigated, and the preliminary examination showed that lactoferrin showed more antibacterial activity compared to the antibiotics, where the diameter of the inhibition zone for *P. aeruginosa* was 13 mm.

**Table 5.** Antimicrobial activity of lactoferrin against representative human pathogenic bacterial strains.

| Bacterial strains | Diameter of inhibition zone (mm) |
|---|---|
| *P. aerugenosae* | 13 |
| *A. baumannii* | 11 |
| *S.aureus* | 14 |
| *S.pyogenes* | 16 |
| **C.S.** $^{(*)}$ ***P-value*** | C.C.=0.067 *P*=0.505 (NS) |

As bacterial growth can be reinstated by the simple addition of exogenous iron in excess of the lactoferrin's chelating capacity, lactoferrins' bacteriostatic action results from their ability to bind to iron and thereby deprive bacteria of this vital nutrient (Arnold et al. 1982). Were the first to describe the bactericidal mechanism of lactoferrins (1977). They showed that human lactoferrin reduced the growth of *Streptococcus* mutans and *Vibrio cholerae* but not *Escherichia coli* in an iron-rich media, and that this antibacterial action was not reversed in the presence of more iron. They hypothesized that human lactoferrin attaches to the surface of bacterial cells based on immunofluorescence investigations.

More research has revealed that human and bovine lactoferrins can directly interact with bacterial cell membranes and bind and release lipopolysaccharides (LPS) from enteric Gram-negative bacilli (Yamauchi et al. 1993, Ellison et al. 1991). Additionally, LPS binds to both human and bovine lactoferrin at the same location (Elass Rochard et al. 1995).

Several mechanisms could account for Lf's antimicrobial effect against Gram-negative bacteria. The first mechanism is that Lf is an iron-binding protein that reduces in the context of microorganisms and scavenges free iron. Thus, a lack of iron hinders *Pseudomonas* from forming biofilms. Patients with cystic fibrosis have a high incidence of biofilm formation, which was suggested as a colonial organization adhesion technique for *Pseudomonas aeruginosa*. Bacteria develop a strong resistance to the host cell defense system and antibiotic therapy as a result of biofilm development (Caraher et al. 2007).

The need for high levels of iron for the formation of biofilms by various bacteria stains is well established. Therefore, it has been proposed that Lf's role as an iron chelator efficiently inhibits biofilm formation through iron sequestration (Weinberg et al. 2004). According to the second process, when lactoferrin binds to lipid A, it disrupts the gram-negative bacteria's outer membrane, causing the release of lipopolysaccharide (LPS), which can eventually cause alterations in the membrane's permeability (Brink et al. 2002). There have been reports of N-terminal Lf receptors being found on several bacteria' surfaces. Due to cell wall breakage caused by the binding of Lf to these receptors, Gram negative bacteria experience cell death (Arnold et al. 1977).

**Synergistic effect of lactoferrin with antibiotics by discoid diffusion methods**

Synergistic effect of lactoferrin and antibiotics resulting in enhanced antibacterial activity; therefore, the development of resistant pathogenic bacteria could be addressed by the simultaneous action of lactoferrin and antibiotics. In this study the synergistic effects of AgNPs with 5 antibiotics against *P. aeruginosa* genes were studied using the disc diffusion method.

**Table 6.** The inhibition zone lactoferrin with and without different types of antibiotics against multidrug resistance bacteria.

| Bacterial strains | *S. pyogenes* | | | *S. aureus* | | *A. baumannii* | | *P. aerugenosae* | | |
|---|---|---|---|---|---|---|---|---|---|---|
| The concentration of L.f | 40 | 20 | 10 | 20 | 10 | 20 | 10 | 40 | 20 | 10 |
| Control | Nil | | | Nil | | Nil | | Nil | | |
| Standard TOB | 17 | | | 17 | | 17 | | 17 | | |
| Sample | 23 | 22 | 25 | 14 | 11 | 19 | 22 | 21 | 17 | 20 |
| Standard Nitro | 18 | | | 18 | | 18 | | 18 | | |
| Sample | 25 | 23 | 21 | 15 | 18 | 21 | 19 | 24 | 23 | 19 |
| Standard Nalid | 16 | | | 16 | | 16 | | 16 | | |
| Sample | 24 | 24 | 22 | 21 | 25 | 30 | 29 | 29 | 31 | 28 |
| Standard Chloram | 17 | | | 17 | | 17 | | 17 | | |
| Sample | 21 | 18 | 20 | 16 | 21 | 14 | 13 | 15 | 24 | 23 |
| Standard Ampic | 15 | | | 15 | | 15 | | 15 | | |
| Sample | 21 | 24 | 25 | 26 | 28 | 26 | 23 | 22 | 24 | 25 |
| **C.S.** (*) ***P-value*** | C.C.=0.213 P=0.096 (NS) | | | C.C.=0.057 P=0.663 (NS) | | C.C.=0.059 P=0.558 (NS) | | C.C.=0.102 P=0.789 (NS) | | |

**Statistical Analysis**

Table 6 shows the normality scores of the dependent variables. Since the significance scores of the Shapiro-Wilk test is less than $p<0.05$, the variables are not normally distributed. Therefore, non-parametric test should be used. Kruskal Wallis H test is used for the group comparison and Mann-Whitney U test is used to understand the difference within the groups.

**Table 7.** Descriptive Statistics of the Variables.

| | Gender | Age | Sample Type | Resistant | Sensitive | Diameter | Bacteria Type |
|---|---|---|---|---|---|---|---|
| N-Valid | 150 | 150 | 150 | 150 | 150 | 150 | 150 |
| Missing | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Std. Deviation | .45 | 18.94 | 1.08 | .086 | .055 | 1.22 | 1.60 |
| Minimum | 1.00 | 16.00 | 1.00 | .30 | .37 | 11.00 | 1.00 |
| Maximum | 2.00 | 90.00 | 4.00 | .57 | .55 | 16.00 | |
| Percentiles 25 | 1.00 | 28.00 | 1.00 | .55 | .38 | 13.00 | 2.00 |
| 20 | 1.00 | 43.50 | 2.00 | .55 | .38 | 13.00 | 4.00 |
| 75 | 2.00 | 58.25 | 2.25 | .57 | .40 | 14.00 | 4.00 |

Table 8 shows the normality scores of the dependent variables. Since the significance scores of the Shapiro-Wilk test is less than $p<0.05$, the variables are not normally distributed. Therefore, non-parametric test should be used. Kruskal Wallis H test is used for the group comparison and Mann-Whitney U test is used to understand the difference within the groups.

**Table 8.** Post-hoc results of the variables based on the resistant scores.

| Sample1 Sample2 | Test statistic | Std. Error | Std. Test statistic | Sig | Adj.Sig, |
|---|---|---|---|---|---|
| Sp-sa | 47.50 | 10.418 | 4.560 | .000 | .000 |
| sp-sa+sp | 47.50 | 15.134 | 3.139 | .002 | .025 |
| sp-ab | 29.00 | 1.378 | 6.399 | .000 | .000 |
| sp-pa | 119.5 | 11.57 | 10.32 | .000 | .000 |
| sp-pa+sp | 199.5 | 15.61 | 7.65 | .000 | .000 |
| sa-sa+sp | .000 | 13.03 | .000 | 1.00 | 1.00 |
| sa-ab | 44.5 | 12.13 | 3.66 | .000 | .004 |
| sa-pa | 72.00 | 8.626 | 8.34 | .000 | .000 |
| sa-pa+sp | 72.00 | 13.57 | 5.30 | .000 | .000 |
| sa+sp-ab | 44.50 | 16.36 | 2.72 | .007 | .098 |
| Sa+sp-pa | 72.50 | 13.96 | 5.16 | .000 | .000 |
| Sa+sp- pa+sp | 72.00 | 17.45 | 4.12 | .000 | .001 |
| ab-pa | 27.50 | 13.14 | 2.09 | .036 | .545 |
| ab_pa+sp | 27.50 | 16.80 | 1.63 | .102 | 1.00 |
| pa-pa+sp | .000 | 14.47 | .000 | 1.00 | 1.00 |

The resistant, sensitive and diameter are used as a dependent variable separately and Kruskal Wallis test for each variable showed that there is a statistically significant difference between the bacterium types. There is the pairwise comparison of the bacteria type considering resistant scores. The significance scores less than 0.05 shows the statistically significant differences.

**Table 9.** Post-hoc results of the variables based on the sensitive scores.

| Sample1 Sample2 | Test statistic | Std. Error | Std. Test statistic | Sig | Adj.Sig |
|---|---|---|---|---|---|
| Ap-sa | -44.50 | 12.14 | -3.664 | .000 | .004 |
| Ap-sa+sp | -44.50 | 16.36 | -2.791 | .007 | .098 |
| Ap-ba | 103.50 | 13.14 | 7.878 | .000 | .000 |
| Ap-ba+sp | 103.50 | 16.80 | 6.160 | .000 | .000 |
| ab-sp | -134.00 | 14.38 | -9.320 | .000 | .000 |
| sa-sa+sp | .000 | 13.03 | .000 | 1.00 | 1.00 |
| sa-ab | 59.50 | 8.62 | 6.840 | .000 | .000 |
| sa-pa+sp | 59.00 | 13.57 | 4.348 | .000 | .000 |
| Sa-sp | -89.50 | 10.42 | -8.591 | .000 | .000 |
| sa+sp-pa | 59.00 | 13.96 | 4.226 | .000 | .000 |
| Sa+sp-pa+sp | 59.00 | 17.45 | 3,380 | .001 | .011 |
| Sa+sp-sp | -89.50 | 15.13 | -5.914 | .000 | .000 |
| Pa-pa+sp | 000 | 14.47 | .000 | 1.00 | 1.00 |
| Pa-sp | -30.50 | 11.57 | -2.636 | .008 | .126 |
| pa-pa+sp | -30.50 | 15.60 | -1.954 | .051 | .760 |

## Conclusion

According to the study's findings, 172 (68.8%) of the samples had bacterial growth that had been grown at an ideal level. The findings revealed that 67 samples of urine yielded 15 isolates of *P. aeruginosa*. Out of 68 wound isolates, there were 19 *P. aeruginosa* isolates, yielding a rate of 27.9%. Moreover, in the current investigation, 5 isolates of *P. aeruginosa* were isolated from 11 blood samples with a proportion of 45.5%.

The findings of the current study indicated that 67 urine samples yielded 1 isolate of *A. baumonii*. 3 *A. baumonii* isolates were found in 68 wound samples. 3 *A. baumonii* isolates were found in 11 blood samples. 26 samples of sputum yielded 6 *A. baumonii* isolates. The findings revealed that 40 *S. aureus* isolates were discovered from 67 urine samples. 25 *S. aureus* isolates were found in 68 wound samples. *S. aureus* was isolated from 1 of 11 blood samples. 26 samples of sputum yielded 12 *S. aureus* isolates. The findings revealed that 11 isolates of *S. pyogenus* were found from 67 urine samples. *S.pyogenus* was isolated from 21 out of 68 wound samples. *Streptococcus pyogenus* was isolated from 2 of 11 blood samples. 26 samples of sputum contained 5 isolates of *S.pyogenus*. *P.aeruginosa* showed total resistance to Amoxicillin/clavulanic acid, Ampicillin, Tetracycline, Cefoxitin, Ceftriaxone, Chloramphenicol and Trimethoprim. A. baumannii showed total resistance to Amoxicillin/clavulanic acid, Ampicillin, Gentamicin, Amikacin, Chloramphenicol, Nalidixic acid, Ciprofloxacin, Trimethoprim and Tobramycin. *S. aureus* showed total sensitive to Nitrofurantoin and Refampicin respectively. *S. pyogenes* showed total resistance to Ampicillin, Tetracycline and Ceftriaxone. Otherwise, *S. pyogenes* showed total sensitive to Penicillin, Cefepime, Vancomycin and Clindamycin. The results of lactoferrin showed that the inhibition zone diameter was (13, 11, 14, 16 mm) for *P. aeruginosa*, *A. baumannii*, *S. aureus* and *S. pyogenes* respectively.

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